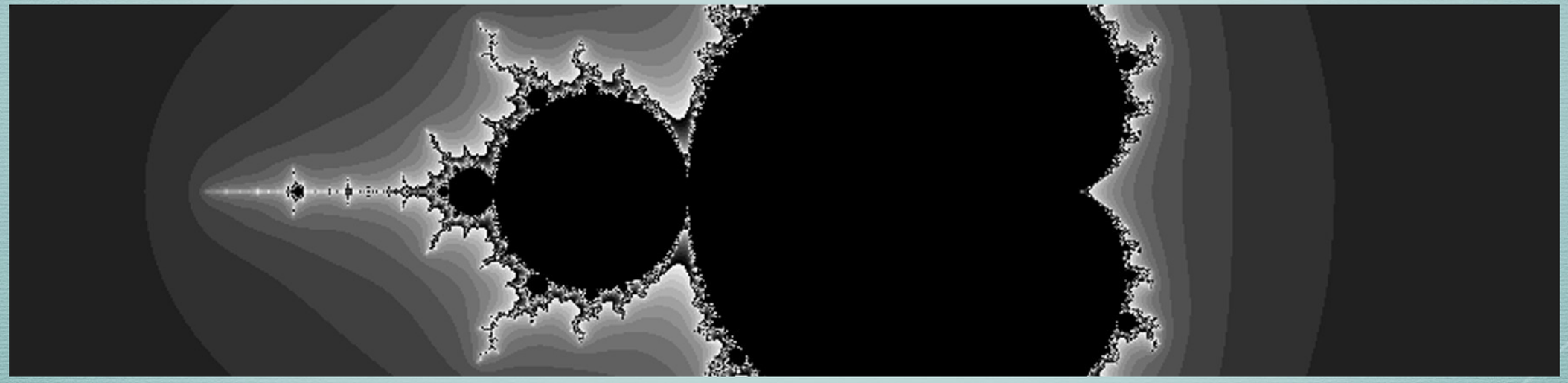
Canvas for the Nature I

Fractal 碎形

碎形在所有的大小尺度下都顯得相似，所以通常被認為是無限複雜的。自然界裡一定程度上類似碎形的事物有雲、山脈、閃電、海岸線、雪片、植物根、多種蔬菜（如花椰菜和西蘭花）和動物的毛皮的圖案等等。

(source:https://zh.wikipedia.org/zh-tw/%E5%88%86%E5%BD%A2)

Mandelbrot set



Outlines

* The power of recursion

* The circles
  * Recursive Tree
  * Pythagorean
  * Tree Koch Curve

HTML Canvas Skeleton code

```
<!DOCTYPE html>
<html><body>
<canvas id="fractal" width="300"
height="300"  style="border:1px solid
#000000;"></canvas>

<script>
  //   取得 canvas
  var canvas =
document.getElem
entById("fractal");
  //   取得用來描繪
  的上下文
  var ctx =
  canvas.getContext
  ("2d");
  //   定義描繪矩形
  的方法
</script>
</body></
html>
```




Monster Curve

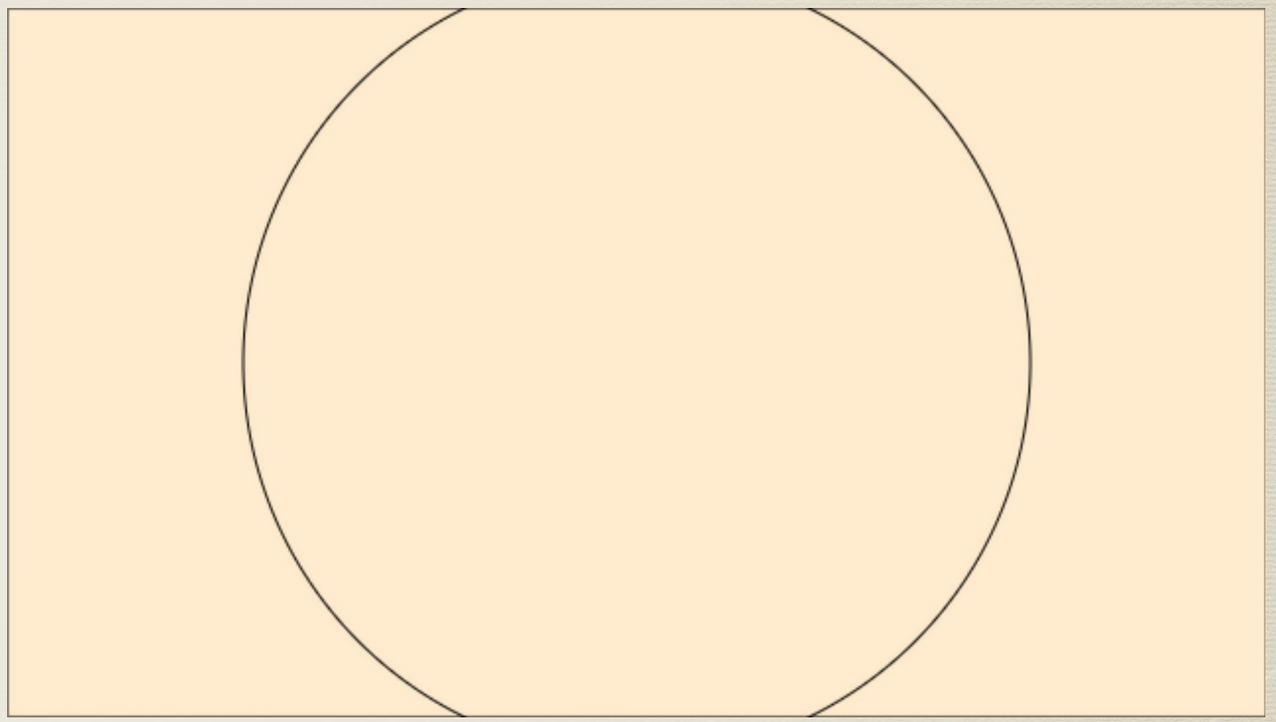
The Power of Recursive for generating Curves

such fractal objects were mathematical monster

Step 1

```
var canvas = document.getElementById("fractal");
    var ctx = canvas.getContext('2d');
    drawCircle(320, 180, 200);
    ctx.strokeRect(0, 0, 640, 360);

    function drawCircle(x, y, r) {
        ctx.beginPath();
        ctx.arc(x, y, r, 0, 2 * Math.PI);
        ctx.stroke();
    }
```



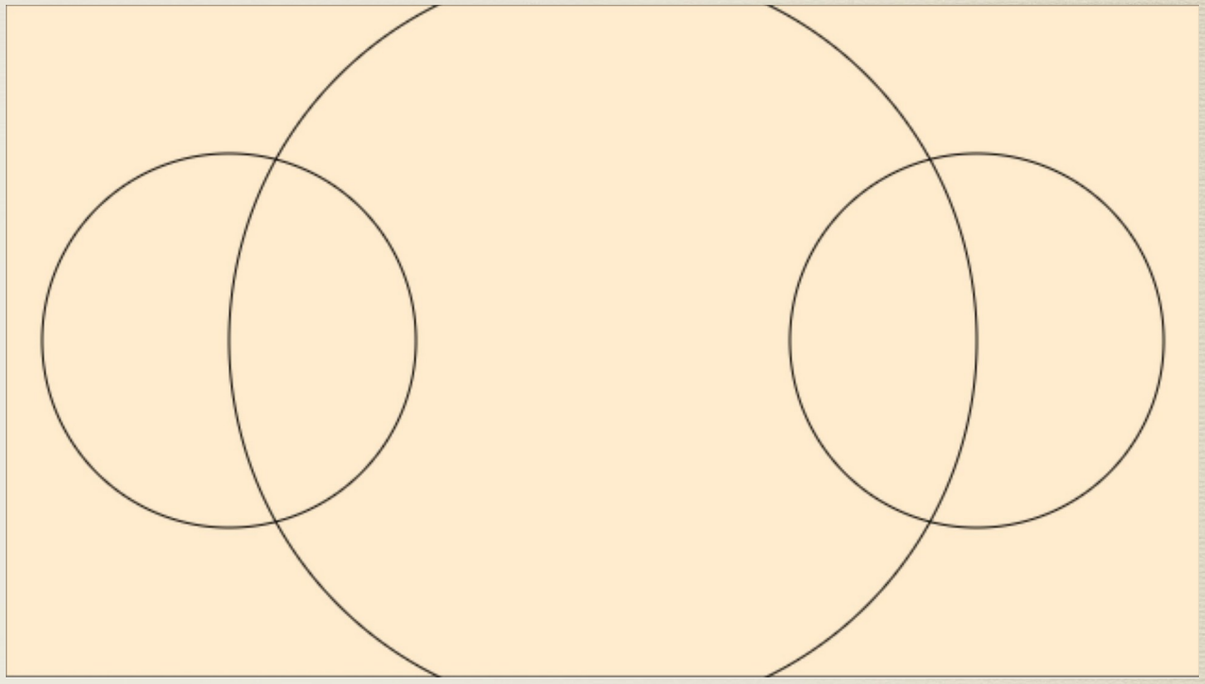
Step 2

```
var canvas = document.getElementById("fractal");
    var ctx = canvas.getContext('2d');
    drawCircle(320, 180, 200);
    ctx.strokeRect(0, 0, 640, 360);

    function drawCircle(x, y, r) {
        ctx.beginPath();
        ctx.arc(x, y, r, 0, 2 * Math.PI);
        ctx.stroke();

        ctx.beginPath();
        ctx.arc(x-r, y, r/2, 0, 2 * Math.PI);
        ctx.stroke();

        ctx.beginPath();
        ctx.arc(x+r, y, r/2, 0, 2 * Math.PI);
        ctx.stroke();
    }
```



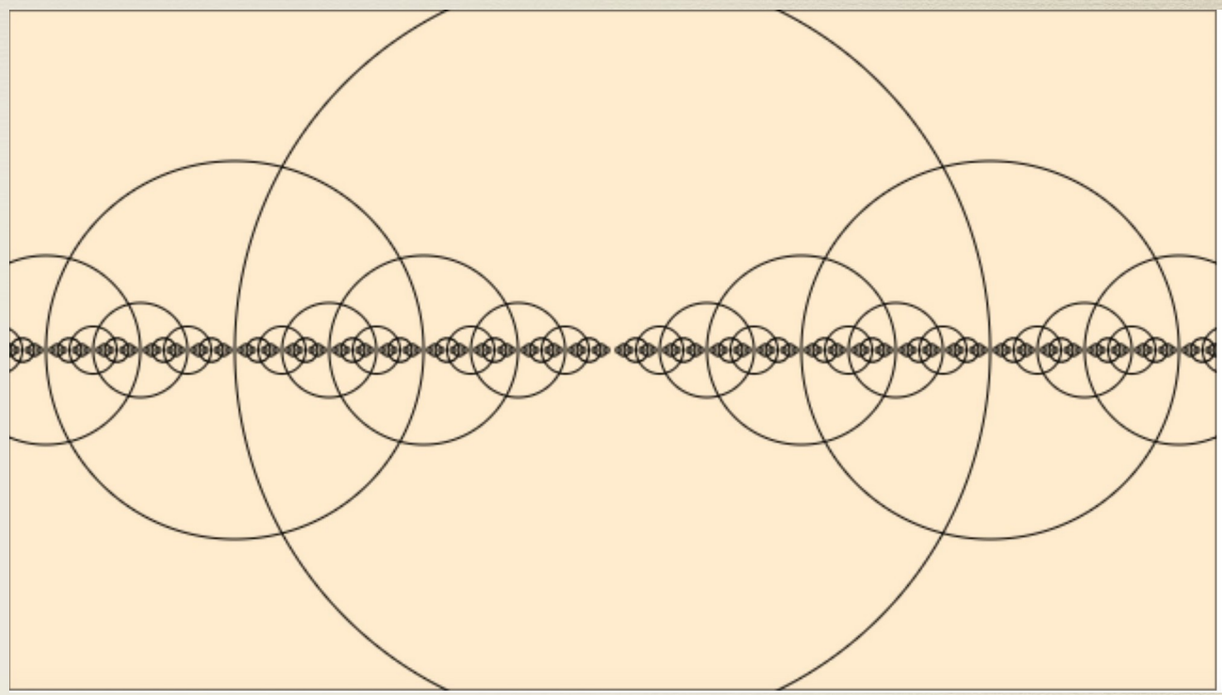
Step 3: Recursive

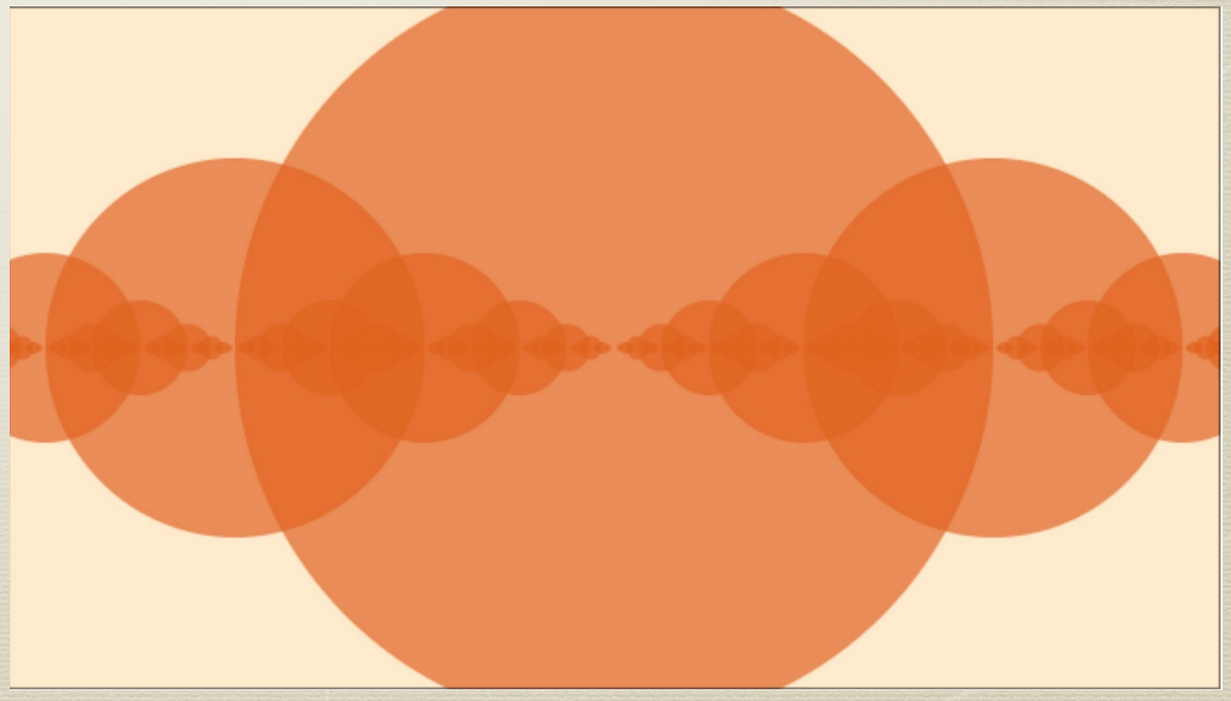
```
function drawCircle(x, y, r) {
        ctx.beginPath();
        ctx.arc(x, y, r, 0, 2 * Math.PI);
        ctx.stroke();

        if (r > 2) {  // condition for drawing

            similarity

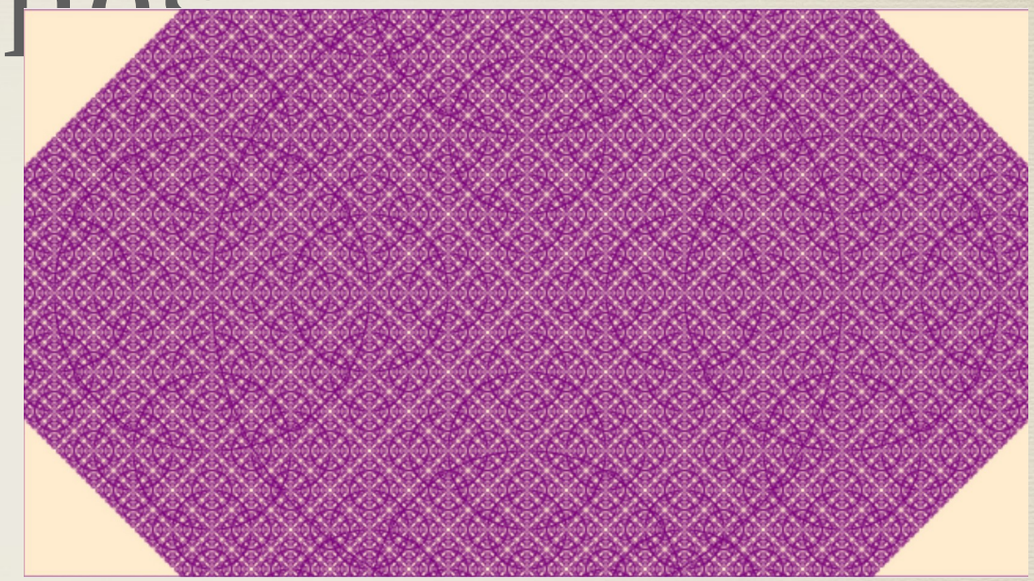
            drawCircle(x + r, y, r / 2);
            drawCircle(x - r, y, r / 2);
        }
    }
```



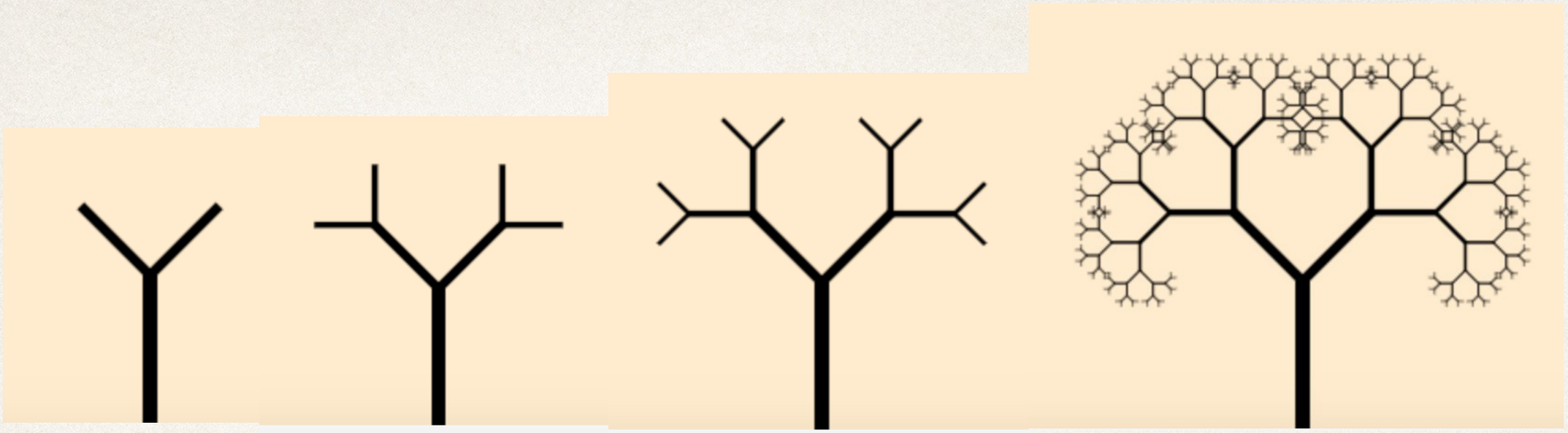


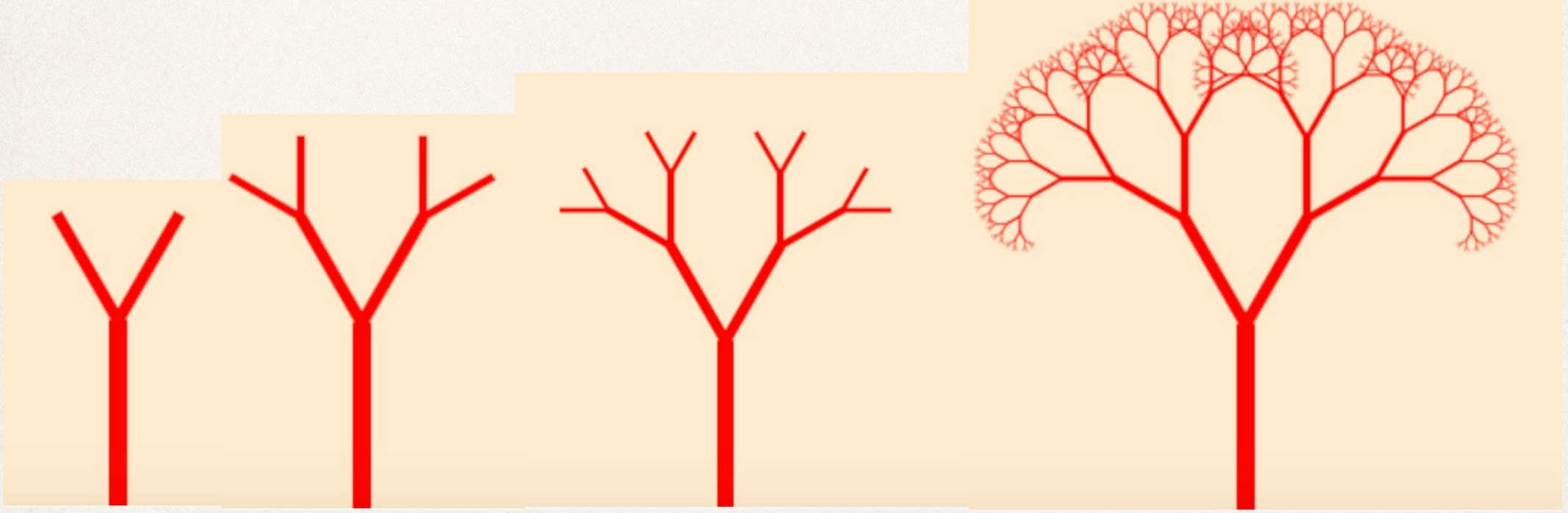
Step 3: add two branches





The Recursive Tree





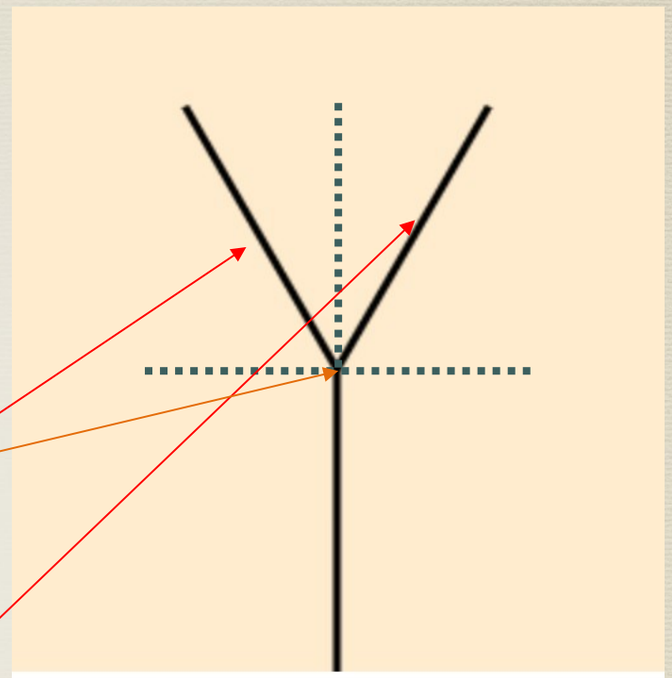
Step 1

```
var canvas = document.getElementById("fractal");
var ctx = canvas.getContext('2d');

ctx.translate(canvas.clientWidth / 2, canvas.clientHeight);

var len = 120;
var theta = 30;
line(0, 0, 0, -len);

ctx.translate(0, -len);
// move coordinate frame to the tree top
ctx.save();
ctx.rotate(Math.PI / 180 * theta);
line(0, 0, 0, -len);
ctx.restore();
ctx.save();
this.ctx.rotate(-Math.PI / 180 * theta);
line(0, 0, 0, -len);
ctx.restore();
```



```
function line(x1, y1, x2, y2,width=3) {

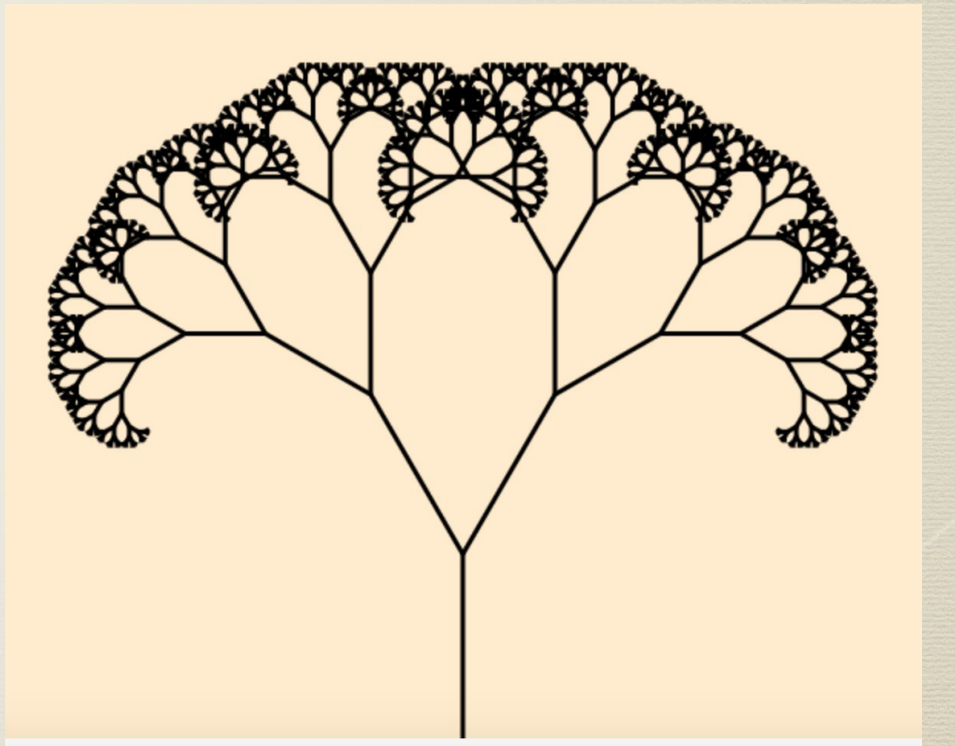
        ctx.strokeStyle = this.color;
        ctx.lineWidth = width;
        ctx.beginPath();
        ctx.moveTo(x1, y1);
        ctx.lineTo(x2, y2);
        ctx.closePath();
        ctx.stroke();

    }
```

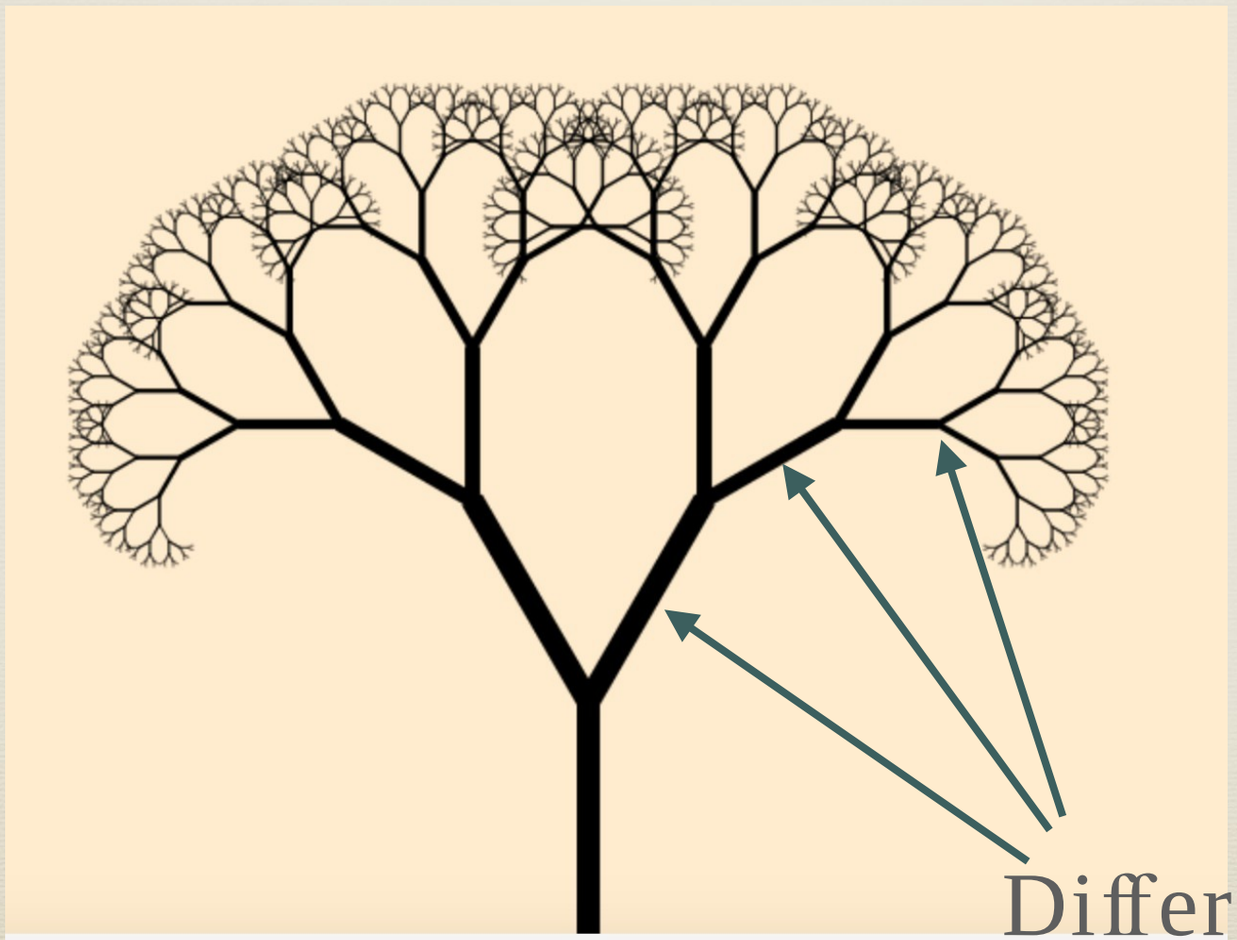
Step 2: draw subtree with branch

```
function branch(len) {

        line(0, 0, 0, -len);
        ctx.translate(0,  -len);
        len *= 0.66;
        if (len > 2) {
            // right branch
            ctx.save();
                ctx.rotate(Math.PI / 180 * theta);
                branch(len);
            ctx.restore();
            // left branch
            ctx.save();
                ctx.rotate(-Math.PI / 180 * theta);
                branch(len);
            ctx.restore();
        }
    }
```



Step 3: draw branch with different trunk size



Step 3: draw branch with different



Different trunk angles

Step4 : makeforest using class

```
var canvas = document.getElementById("fractal");
var ctx = canvas.getContext('2d');

ctx.translate(canvas.clientWidth / 2, canvas.clientHeight);

var len = 120;
var theta = 30;

branch();
```

加上 for 迴圈，並重置每棵樹的起點
數高、顏色、樹枝角度寬度皆可利用 Math.random() 變化

